

Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 12

Deadline: next Wednesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems.

Problem 1. For each of the following generator matrices, say how many errors the corresponding code detects and how many errors it corrects:

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}; \quad \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

Problem 2. Let $f : \mathbb{B}^3 \rightarrow \mathbb{B}^9$ be the coding function given by

$$f(abc) = abcabc\bar{a}\bar{b}\bar{c},$$

where $\bar{x} = 1$ if $x = 0$ and $\bar{x} = 0$ if $x = 1$.

- (i) List the eight codewords of f .
- (ii) Show that f does not give a group code.
- (iii) How many errors does f detect and how many errors does it correct?

Problem 3. Give the complete coset decoding table for the code given by the generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

Problem 4. Let d denote the (Hamming) distance on $\mathbb{B}^n$. Prove the following:

- (a) $d(x, y) \geq 0$ for all $x, y \in \mathbb{B}^n$.
- (b) $d(x, y) = 0$ if and only if $x = y$.
- (c) $d(x, y) = d(y, x)$ for all $x, y \in \mathbb{B}^n$.
- (d) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in \mathbb{B}^n$.